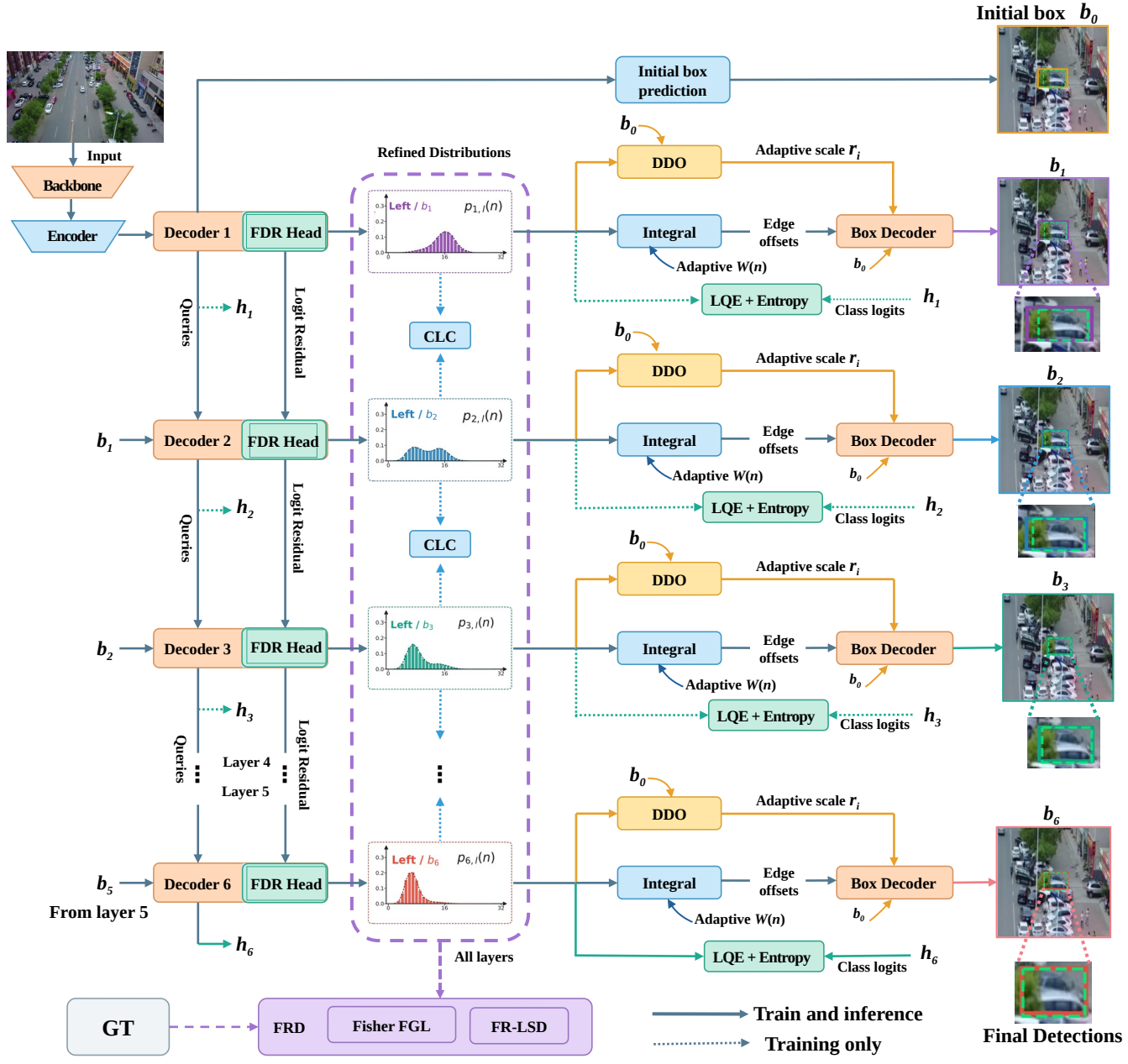
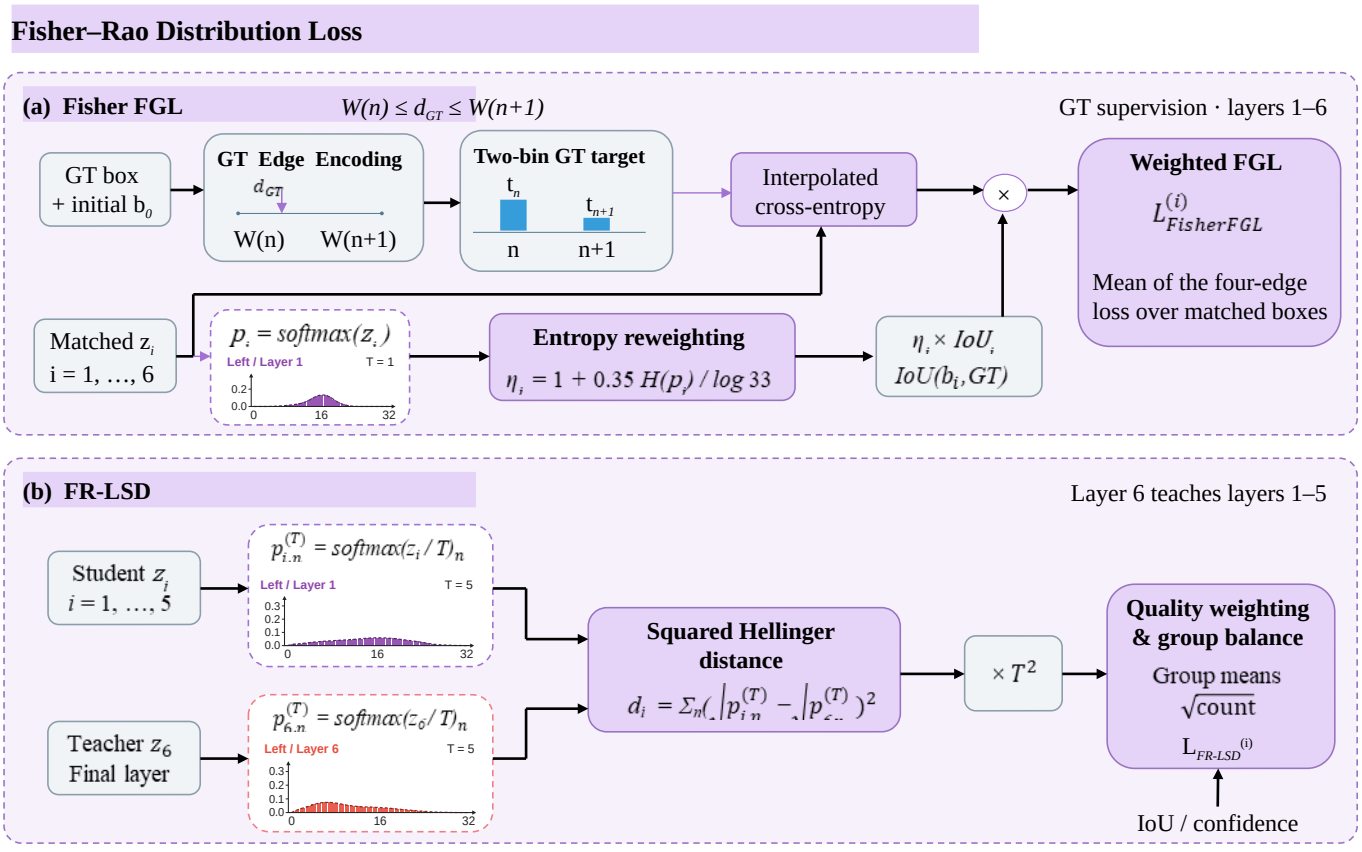
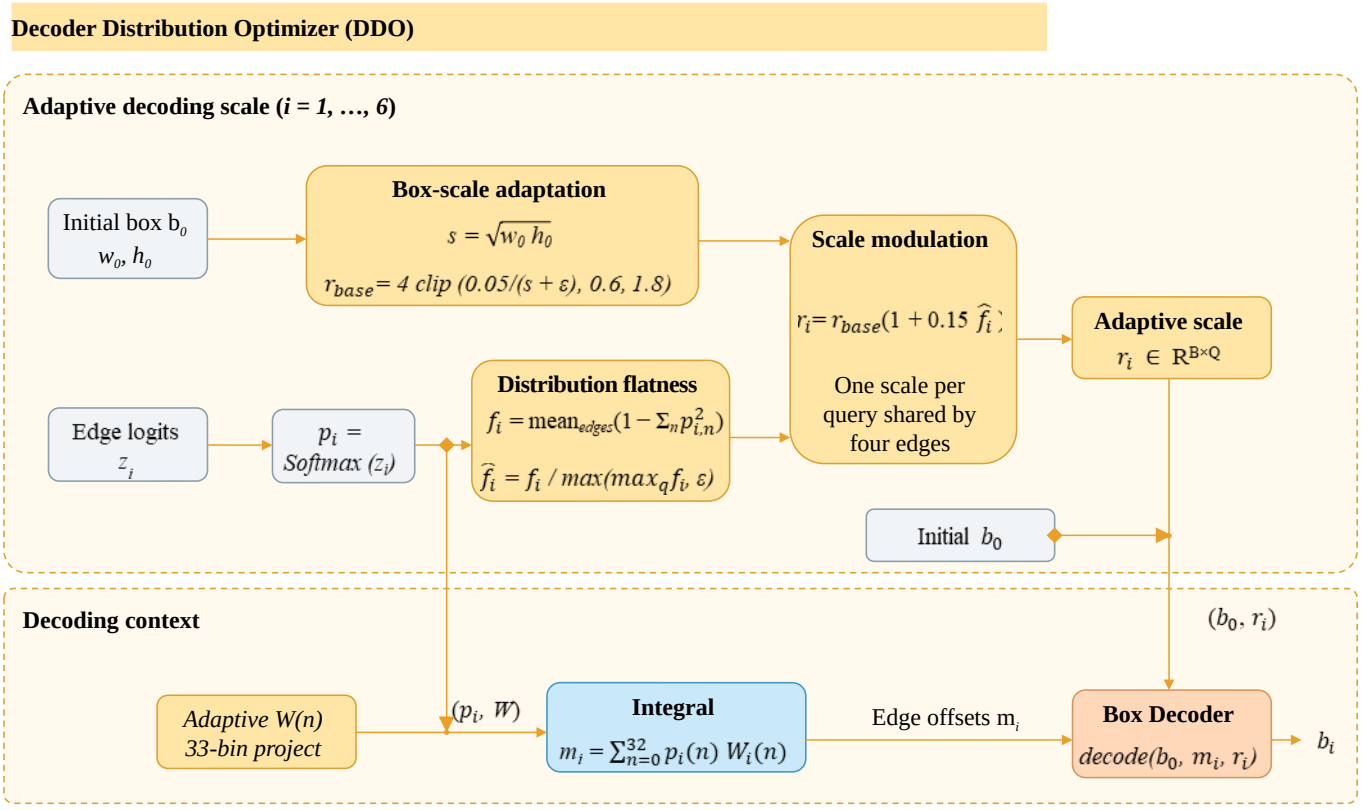
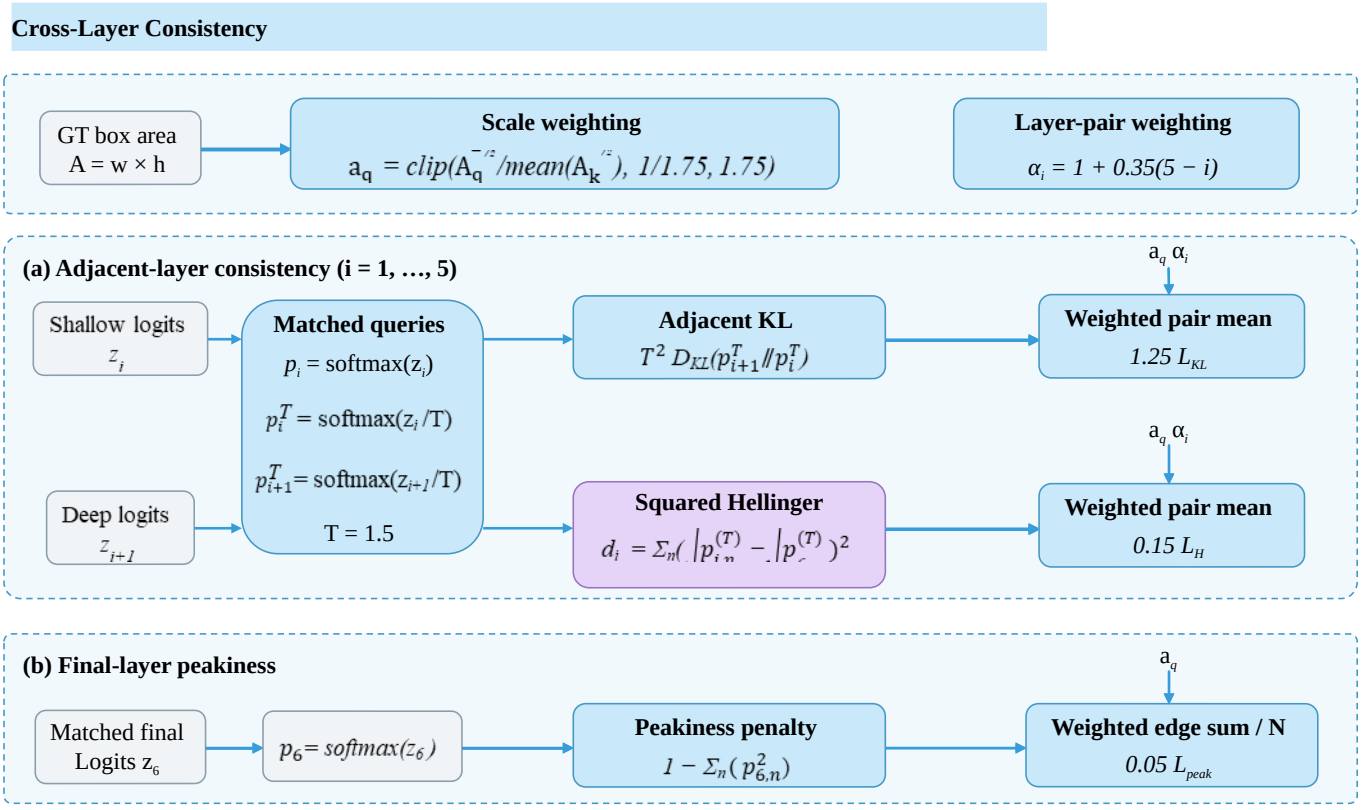
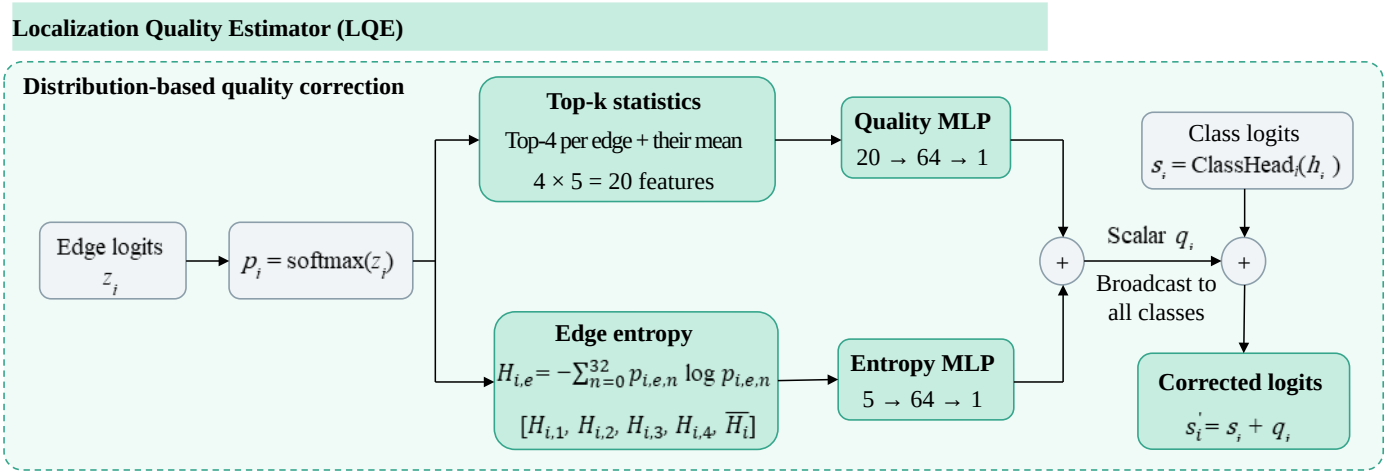

Initial box b_0
Input
Backbone
Encoder
Initial box prediction
Refined Distributions
Decoder 1
FDR Head
Queries
h_1
Logit Residual
b_1
Decoder 2
FDR Head
Queries
h_2
Logit Residual
b_2
Decoder 3
FDR Head
Queries
h_3
Layer 4
Layer 5
Logit Residual
b_5
From layer 5
Decoder 6
FDR Head
h_6
Left / b_1 $p_{1,i}(n)$
Left / b_2 $p_{2,i}(n)$
Left / b_3 $p_{3,i}(n)$
Left / b_6 $p_{6,i}(n)$
CLC
CLC
All layers
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_1
b_1
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_2
b_2
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_3
b_3
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_6
b_6
Final Detections
GT
FRD
Fisher FGL
FR-LSD
Train and inference
Training only


Fisher–Rao Distribution Loss
(a) Fisher FGL
W(n) ≤ d_GT ≤ W(n+1)
GT supervision · layers 1–6
GT box + initial b_0
GT Edge Encoding
d_GT
W(n)
W(n+1)
Two-bin GT target
t_n
t_n+1
n
n+1
Interpolated cross-entropy
×
Weighted FGL
L^(i)_FisherFGL
Mean of the four-edge loss over matched boxes
Matched z_i
i = 1, …, 6
p_i = softmax(z_i)
Left / Layer 1
T = 1
0.2
0.0
0
16
32
Entropy reweighting
η_i = 1 + 0.35 H(p_i) / log 33
η_i × IoU_i
IoU(b_i, GT)
(b) FR-LSD
Layer 6 teaches layers 1–5
Student z_i
i = 1, …, 5
p^(T)_i,n = softmax(z_i / T)_n
Left / Layer 1
T = 5
0.3
0.2
0.1
0.0
0
16
32
Squared Hellinger distance
d_i = Σ_n(√p^(T)_i,n − √p^(T)_6,n)^2
× T^2
Quality weighting & group balance
Group means
√count
L_FR-LSD^(i)
IoU / confidence
Teacher z_6
Final layer
p^(T)_6,n = softmax(z_6 / T)_n
Left / Layer 6
T = 5
0.3
0.2
0.1
0.0
0
16
32


Decoder Distribution Optimizer (DDO)
Adaptive decoding scale (i = 1, …, 6)
Initial box b0
w0, h0
Box-scale adaptation
s = √w0 h0
rbase = 4 clip (0.05/(s + ε), 0.6, 1.8)
Scale modulation
ri = rbase(1 + 0.15 f̂i)
One scale per query shared by four edges
Adaptive scale
ri ∈ RB×Q
Edge logits
zi
pi = Softmax (zi)
Distribution flatness
fi = mean edges(1 − Σn p2i,n)
f̂i = fi / max(maxq fi, ε)
Initial b0
Decoding context
(b0, ri)
Adaptive W(n)
33-bin project
(pi, W)
Integral
mi = Σ32n=0 pi(n) Wi(n)
Edge offsets mi
Box Decoder
decode(b0, mi, ri)
bi


Cross-Layer Consistency
GT box area
A = w × h
Scale weighting
$a_q = clip(A_q^{-/2}/mean(A_k^{-/2}), 1/1.75, 1.75)$
Layer-pair weighting
$\alpha_i = 1 + 0.35(5 - i)$
(a) Adjacent-layer consistency (i = 1, …, 5)
a_q α_i
Shallow logits
z_i
Matched queries
$p_i = \text{softmax}(z_i)$
$p_i^T = \text{softmax}(z_i/T)$
$p_{i+1}^T = \text{softmax}(z_{i+1}/T)$
T = 1.5
Deep logits
z_{i+1}
Adjacent KL
$T^2 D_{KL}(p_{i+1}^T \| p_i^T)$
Weighted pair mean
$1.25\ L_{KL}$
a_q α_i
Squared Hellinger
$d_i = \Sigma_n(|p_{i,n}^{(T)} - |p^{(T)}|)^2$
Weighted pair mean
$0.15\ L_H$
(b) Final-layer peakiness
a_q
Matched final
Logits z_6
$p_6 = softmax(z_6)$
Peakiness penalty
$1 - \Sigma_n(p_{6,n}^2)$
Weighted edge sum / N
$0.05\ L_{peak}$


Localization Quality Estimator (LQE)
Distribution-based quality correction
Edge logits
z_i
$p_i = \text{softmax}(z_i)$
Top-k statistics
Top-4 per edge + their mean
4 × 5 = 20 features
Quality MLP
20 → 64 → 1
Edge entropy
$H_{i,e} = -\sum_{n=0}^{32} p_{i,e,n} \log p_{i,e,n}$
$[H_{i,1}, H_{i,2}, H_{i,3}, H_{i,4}, \overline{H_i}]$
Entropy MLP
5 → 64 → 1
Class logits
$s_i = \text{ClassHead}_i(h_i)$
Scalar q_i
Broadcast to all classes
+
+
Corrected logits
$s_i' = s_i + q_i$